

JavaScript Mastery Study Plan (Frontend + React + Node.js + MERN)

Phase 1 — Core JavaScript

Variables (var, let, const); Data Types; Type Conversion & Coercion; Operators; Template Literals; User Input; Conditional Statements; Loops; Break & Continue

Phase 2 — Functions

Function Declaration, Expression, Arrow Functions, Parameters & Arguments, Return, Default Parameters, Rest, Spread, Callback Functions, Higher Order Functions, IIFE, Pure vs Impure Functions, Recursion

Phase 3 — Arrays

Basic methods: push, pop, shift, unshift, slice, splice, concat, join, reverse, sort. Advanced: map, filter, reduce, find, findIndex, some, every, includes, flat, flatMap.

Phase 4 — Strings

length, trim, toUpperCase, toLowerCase, slice, substring, replace, replaceAll, split, includes, startsWith, endsWith.

Phase 5 — Objects

Creating objects, properties, nested objects, methods, this, destructuring, spread, Object.keys(), values(), entries(), assign(), freeze().

Phase 6 — ES6+

let/const, Arrow Functions, Template Literals, Destructuring, Spread, Rest, Enhanced Object Literals, Optional Chaining, Nullish Coalescing, Modules.

Phase 7 — Scope & Execution

Global, Function, Block, Lexical Scope, Hoisting, TDZ, Execution Context, Call Stack.

Phase 8 — Advanced Functions

Closures, Currying, call(), apply(), bind(), Function Borrowing.

Phase 9 — DOM

Introduction, Selecting, Modifying, Styling, Creating, Removing, Traversal.

Phase 10 — Events

Listeners, Event Object, Bubbling, Capturing, Delegation, preventDefault(), stopPropagation().

Phase 11 — Asynchronous JavaScript

Sync vs Async, Callback Hell, Promises, Promise.all(), allSettled(), race(), any().

Phase 12 — Async/Await

async, await, try/catch, Error Handling, Fetch with async/await.

Phase 13 — APIs & Networking

HTTP, REST, JSON, Fetch API, GET/POST/PUT/PATCH/DELETE, Headers, Request Body, Authentication Basics.

Phase 14 — Browser Storage

localStorage, sessionStorage, Cookies, JSON.stringify(), JSON.parse().

Phase 15 — Error Handling

try, catch, finally, throw, Custom Errors.

Phase 16 — OOP

Constructor Functions, Prototype, Prototype Chain, Classes, Encapsulation, Inheritance, Polymorphism, Static Methods.

Phase 17 — JavaScript Internals

Event Loop, Web APIs, Microtasks, Macrotasks, Memory Management, Garbage Collection.

Phase 18 — Performance

Debouncing, Throttling.

Phase 19 — Modules

ES Modules, Named Export, Default Export, Dynamic Import.

Phase 20 — Functional Programming

First-Class Functions, Higher Order Functions, Pure Functions, Immutability, Function Composition.

Phase 21 — Interview Revision

var vs let vs const, == vs ===, Hoisting, Scope, Closures, this, call/apply/bind, Prototype, Event Loop, Promise vs async/await, Debounce vs Throttle, Shallow vs Deep Copy.

Phase 22 — Real MERN Topics

Form Validation, API Integration, JWT, Cookies, Local Storage, Authentication Flow, File Upload, Pagination, Search, Filter, Infinite Scroll, Lazy Loading.

Phase 23 — Master Projects

Beginner: Calculator, Clock, Stopwatch, Counter. Intermediate: Todo, Weather, Notes, Quiz, Expense Tracker, GitHub Finder, Image Search, Infinite Scroll, Drag & Drop. Advanced: CRUD + Auth, Blog CMS, Chat App, Video Calling, Collaboration App, E-commerce, JWT Auth, Admin Dashboard, Full MERN E-commerce.

Estimated Timeline: 16–20 weeks with 2–3 hours/day of consistent study and project practice.